# 数组计数法2

## 人员

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## 上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/3297

**王向东老师周六一点半C++数组计数法**

刷新

| # | 用户名 | 姓名 | 编程分 | 时间 | A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | chenzhiheng | 陈之恒 | 800 | 43 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 |
| 2 | suihaochen | 隋溟琛 | 800 | 44 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 |
| 3 | zhanggao | 张高 | 800 | 55 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 |
| 4 | zangchenyun | 臧晨云 | 700 | 45 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | |
| 5 | huojiaming | 霍佳铭 | 700 | 45 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | |
| 6 | chuxingyi | 初星熠 | 700 | 47 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | |
| 7 | liukeyang | 刘克阳 | 600 | 40 | 100 | 100 | 100 | 100 | 100 | 100 | | |
| 8 | liuzichen123 | 刘子郴 | 600 | 41 | 100 | 100 | 100 | 100 | 100 | 100 | | |
| 9 | hepeiyu | 何佩瑜 | 600 | 50 | 100 | 100 | 100 | 100 | 100 | 100 | | |
| 10 | liuyinan | 刘奕男 | 600 | 89 | 100 | 100 | 100 | 100 | 100 | 100 | | |
| 11 | liujiahao | 刘嘉浩 | 500 | 2 | 100 | 100 | 100 | 100 | 100 | | | |
| 12 | linyuming | 林钰明 | 500 | 4 | 100 | 100 | 100 | 100 | 100 | | | |
| 13 | yuhaohua | 于昊华 | 500 | 5 | 100 | 100 | 100 | 100 | 100 | | | |
| 14 | lijingxuan | 李景翾 | 500 | 9 | 100 | 100 | 100 | 100 | 100 | | | |

## 作业

https://cppoj.kids123code.com/contest/3406 (课上讲了 A ~ E 题, 作业是 F 题必做, G H 选做)

## 课堂表现

同学们课上做题表现整体都比较好, 其中 C 题有几位同学做的稍慢, 课下可以再复盘一下这道题目。

## 课堂内容

**数字出现次数**

```
#include <bits/stdc++.h>

using namespace std;

int cnt[25];

int main()
{
  int x;
  for (int i = 1; i <= 50; ++i) {
```

```
    cin >> x;
    cnt[x]++;
  }

  int maxx = 0;
  for (int i = 0; i <= 19; i++) {
    if (cnt[i] > maxx) {
      maxx = cnt[i];
    }
  }

  cout << maxx << endl;
  return 0;
}
```

### 去除重复数字

```
#include <bits/stdc++.h>

using namespace std;

int a[100], cnt[1005];

int main()
{
  int n;
  cin >> n;
  for (int i = 1; i <= n; ++i) {
    cin >> a[i];
    cnt[a[i]]++;
  }

  int c = 0;
  for (int i = 0; i <= 1000; i++) {
    if (cnt[i] > 0) c++;
  }

  cout << c << endl;
  for (int i = 0; i <= 1000; i++) {
    if (cnt[i] > 0) {
      cout << i << endl;
    }
  }
  return 0;
}
```

### 排名第二高的成绩

```
#include <bits/stdc++.h>

using namespace std;

int a[100], cnt[105];

int main()
{
  int n;
  cin >> n;
  for (int i = 1; i <= n; ++i) {
    cin >> a[i];
    cnt[a[i]]++;
  }

  int c = 0;
  for (int i = 100; i >= 0; i--) {
    if (cnt[i] > 0) {
      c++;
      if (c == 2) {
        cout << i << endl;
        return 0;
      }
    }
  }
  return 0;
}
```

**找筷子**

```
#include <bits/stdc++.h>

using namespace std;

int a[10005], cnt[1005];

int main()
{
  int n;
  cin >> n;
  for (int i = 1; i <= n; ++i) {
    cin >> a[i];
    cnt[a[i]]++;
  }

  for (int i = 1; i <= 1000; i++) {
    if (cnt[i]%2 == 1) {
      cout << i << endl;
    }
  }
```

```
  return 0;
}
```

### 选班委

```
#include <bits/stdc++.h>

using namespace std;

int a[2005], cnt[205];

int main()
{
  int n, m;
  cin >> n >> m;
  for (int i = 1; i <= m; ++i) {
    cin >> a[i];
    cnt[a[i]]++;
  }

  int maxx = 0;
  for (int i = 1; i <= n; ++i) {
    if (cnt[i] > maxx) {
      maxx = cnt[i];
    }
  }

  for (int i = 1; i <= n; i++) {
    if (cnt[i] == maxx) {
      cout << i << endl;
      return 0;
    }
  }
  return 0;
}
```

### 缺失的数字

```
#include <bits/stdc++.h>

using namespace std;

int a[100005], cnt[100005];

int main()
{
  int n;
  cin >> n;
```

```
  for (int i = 1; i <= n-2; ++i) {
    cin >> a[i];
    cnt[a[i]]++;
  }

  for (int i = 1; i <= n; i++) {
    if (cnt[i] == 0) {
      cout << i << " ";
    }
  }
  return 0;
}
```

### 粉碎数字

```
#include <bits/stdc++.h>

using namespace std;

int a[1005], cnt[15];

int main()
{
  int n, t;
  cin >> n;
  for (int i = 1; i <= n; ++i) {
    cin >> a[i];
    while (a[i] > 0) {
      t = a[i]%10;
      cnt[t]++;
      a[i] /= 10;
    }
  }

  for (int i = 9; i >= 0; i--) {
    for (int j = 1; j <= cnt[i]; j++) {
      cout << i;
    }
  }
  return 0;
}
```

### 扑克牌组合

```
#include <bits/stdc++.h>

using namespace std;
```

```
int a[55], cnt[30];

int main()
{
  int n;
  cin >> n;
  for (int i = 1; i <= n; ++i) {
    cin >> a[i];
  }

  for (int i = 1; i <= n; i++) {
    for (int j = i+1; j <= n; j++) {
      cnt[a[i]+a[j]]++;
    }
  }

  int c = 0;
  for (int i = 1; i <= 26; i++) {
    if (cnt[i] > 0) {
      c++;
    }
  }

  cout << c << endl;
  for (int i = 1; i <= 26; i++) {
    if (cnt[i] > 0) {
      cout << i << " ";
    }
  }
  return 0;
}
```